

Erdős Problem #768

EP-768

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $A \subset \mathbb{N}$ be the set of n such that for every prime $p \mid n$ there exists some $d \mid n$ with $d > 1$ such that $d \equiv 1 \pmod{p}$. Is it true that there exists some constant $c > 0$ such that for all large N $\frac{|A \cap [1, N]|}{N} = \exp(-(c + o(1))\sqrt{\log N} \log \log N)$.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>